Name: _____

YEAR 11 SPECIALIST MATHS INVESTIGATIVE TASK 4

Trigonometry (non-CAS)

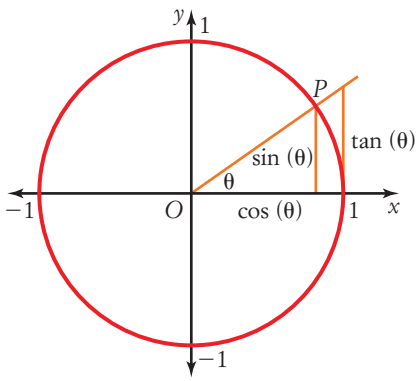
Prerequisites

Knowledge of

- All trigonometric identities and formulae
- Trigonometric ratios of special angles, multiples of $\frac{\pi}{4}$ and $\frac{\pi}{6}$
- Solving polynomial equations (mainly quadratic) exactly

[Total: 80 marks]

1 Consider the unit circle.



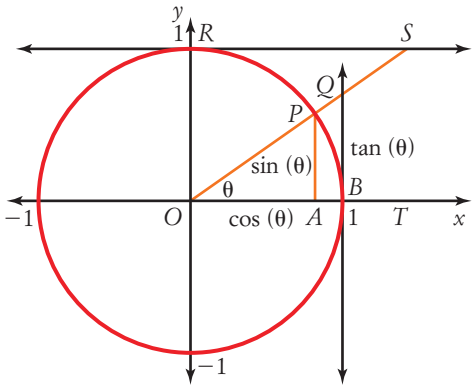
a Explain why $\sin^2(\theta) + \cos^2(\theta) = 1$.

b Write this identity with $\cos^2(\theta)$ as the subject. Use the difference of perfect squares to create a new identity.

c Use the result in **b** to prove that $\frac{\cos(\theta)}{1 - \sin(\theta)} = \frac{1 + \sin(\theta)}{\cos(\theta)}$.

For what values of θ is this identity true?

d Use similar triangles in the extended unit circle diagram below to find a length associated with the unit circle that corresponds to each of the functions sec (θ), cosec (θ), and cot (θ).



[2 + 2 + 4 + 6 = 14 marks]

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2 Determine the exact values of the following, showing all steps. Irrational fractions should be presents with rational denominators.

a $\sec(45°)$

b $\cot\left(\frac{\pi}{6}\right)$

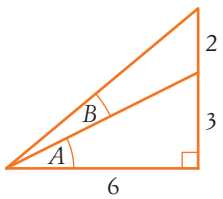
c $\sin\left(\frac{\pi}{12}\right)$

d $\dfrac{2\tan\left(\frac{3\pi}{8}\right)}{1-\tan^2\left(\frac{3\pi}{8}\right)}$

e $\tan\left(\frac{3\pi}{8}\right)$ (using your result in part **d**)

f $\tan(B)$ (without the use of a calculator)

Hint: Let the large angle be $\angle C = \angle A + \angle B$



g $\sin(75°) - \sin(15°)$

h $\cos(85°) + \cos(155°) + \cos(35°)$

i $2\sin(15°)\sin(75°)$

[2 + 2 + 3 + 2 + 6 + 3 + 2 + 2 + 2 = 24 marks]

3 Prove the following trigonometric identities.

a $\dfrac{\cot(x)}{1+\cot(x)} = \dfrac{1}{1+\tan(x)}$

b $\dfrac{1+\sin(P)-\cos^2(P)}{\cos(P)[1+\sin(P)]} = \tan(P)$

c $\dfrac{\sin(\theta)+\sin(2\theta)+\sin(3\theta)}{\cos(\theta)+\cos(2\theta)+\cos(3\theta)} = \tan(2\theta)$

d $\dfrac{\cos(y)+\sin(y)}{\cos(y)-\sin(y)} = \tan(2y)+\sec(2y)$

[2 + 4 + 5 + 6 = 17 marks]

4 a Is it possible to express $\sin(4\theta)$ in terms of $\sin(\theta)$ only without using surds?

b Prove that $\sin(5\theta) = 5\sin(\theta) - 20\sin^3(\theta) + 16\sin^5(\theta)$

c Let $x = \sin\left(\frac{\pi}{10}\right)$.

i Use the result in **b** to find an equation in x.

ii Convert this to an expression which is equal to 0.

iii Given that $16x^5 - 20x^3 + 5x - 1 = (x-1)(4x^2+2x-1)^2$, find a surd expression for $\sin\left(\frac{\pi}{10}\right)$.

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d For which values of n do you think $\sin(n\theta)$ can be expressed in terms of $\sin(\theta)$ only without using surds?

e Use mathematical induction to prove the conjecture 'For odd values of n, $\sin(n\theta)$ can be expressed in terms of odd powers of $\sin(\theta)$ and $\cos(n\theta)$ can be expressed in terms of odd powers of $\cos(\theta)$.'

HINT: Let any odd number be $2n - 1$. The conjecture then would be equivalent to:

$\sin[(2n - 1)\theta] = f[\sin(\theta)]$ and $\cos[(2n - 1)\theta] = g[\cos(\theta)]$

where $f[\sin(\theta)]$ and $g[\cos(\theta)]$ are polynomials in odd powers of $\sin(\theta)$ and $\cos(\theta)$ respectively.

[3 + 5 + 9 + 1 + 7 = 25 marks]

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